

3(a)(i)	B-lymphocyte / plasma cell ; A B-cell / white blood cell / lymphocyte / leucocyte / R beta / β , cells R T-lymphocyte / macrophage / neutrophil	1
3(a)(ii)	<i>three from</i> 1 antibody complementary (shape) to antigen ; I matching shape 2 <i>idea that <u>variable regions</u> different for each antibody ;</i> 3 (specific) antigen / H / N, binding sites (on antibody) ; <i>if mp 2 and 3 not gained, allow one mark for antigen binding regions</i> 4 sequence of amino acids / primary structure, varies between antibodies / gives specificity ; 5 <i>ref. to</i> difference in, primary structure / amino acid sequence, leading to different, tertiary structure / 3-D shape ; 6 <i>ref. to</i> R group interactions between antibody and antigen ;	I ref. to receptor 3
3(b)(i)	<i>Vibrio cholerae</i> ; I bacterium	1
3(b)(ii)	faecal–oral route ;; <i>if mode of transmission not named, allow description</i> <i>infected person</i> <u>faeces</u> / <u>sewage</u> , contaminates, (drinking) water / cooking utensils / vegetable plots / crops / food A diarrhoea for <i>faeces</i> R (human) waste unqualified or <i>ref. to</i> houseflies landing on contaminated faces ; <i>uninfected person</i> eating contaminated food / using contaminated utensils / drinking contaminated water ; A bacteria enters water in context of drinking R infected food or water I handling contaminated food <i>need ref. to contaminated once only</i>	2

Question	Answer	Marks
3(c)(i)	cell wall, breaks / damaged / AW A cell wall becomes thinner or cell wall weakened ; <i>plus one from</i> detail of change to cell wall e.g. cross links break down, quantity of murein / peptidoglycan, decreases or cell membrane, breaks / AW or increase in water in cell (so increase in turgor pressure) / AW ;	2
3(c)(ii)	<i>two from</i> infect only, <i>V. cholerae</i> / cholera bacteria AW or do not, infect human cells / infect humans ; A do not harm human cells do not give side effects / allergic responses, (to humans) ; able to replicate inside <i>V. cholerae</i> (to produce more bacteriophage for treatment) ; <i>ref. to</i> remaining, active / infective / AW, with delivery method used / within gut ;	2
3(c)(iii)	<i>antibiotics</i> only effective against bacteria / not effective against viruses or do not act on structures possessed by, (measles) virus / <i>Morbillivirus</i> A named structure e.g. protein coat or act on structures that viruses do not have A named structure e.g. murein cell wall or only act on, living / growing, cells or may not be able to reach virus inside (host) cell ; AW	1